ECE 310 Fall 2026
Lecture 9: Transfer functions and LTI system response: Part 1

Corey Snyder

University of Illinois Urbana-Champaign

September 14th, 2026



Learning Objectives

- Compute the response of an LTI system to an input signal using the z-transform.
- Write the transfer function of an LTI system described by an LCCDE.
- Identify FIR and IIR systems based on characteristics of an LCCDE and its transfer function.



LTI System Response and the z-transform

For an LTI system with impulse response $h[n]$, let $H(z)$ denote the system's transfer function. We know thus far that

$$y[n] = x[n] * h[n].$$

Now, we may express the input-output relationship of LTI systems a second way using the convolution property of the z-transform:

$$Y(z) = X(z)H(z), \ \text{ROC} \ = \ \text{at least} \ R_x \cap R_h.$$

An important consequence of the convolution property is that we see another way to express the transfer function of an LTI system:

$$H(z) = \frac{Y(z)}{X(z)}.$$



Transfer Functions and LCCDEs

Recall from previous lectures how we defined general Linear Constant Coefficient Difference Equations (LCCDE):

$$y[n] + \sum_{k=1}^{N} a_k y[n-k] = \sum_{k=0}^{M-1} b_k x[n-k].$$

What happens if we take the z-transform of both sides?



Exercise 1: Finding Impulse Response

Compute the impulse response of the following causal LCCDE:

$$y[n] = \frac{1}{3}y[n-1] + x[n]$$



Exercise 2: Finding Impulse Response

Compute the impulse response of the following causal LCCDE:

$$y[n] = \frac{1}{3}y[n-1] + x[n] - 3x[n-1] + x[n-2]$$



Exercise 3: Finding Impulse Response

Compute the impulse response of the following causal LCCDE:

$$y[n] = \frac{1}{2}y[n-1] + \frac{3}{16}y[n-2] + x[n] - 3x[n-1]$$



Characterizing Transfer Functions

